

UDEE Practice Test 08: Answer Key

30 questions · Section A Figural, Section B Observation, Section C Logic-Based Reasoning

Review rules. Read the explanation for every question you missed AND every question you guessed correctly. Pay extra attention to Q21 (necessary condition direction), Q28 (quantifier syllogism), and Q30 (new 4-statement self-reference).

Answer Grid

1. C	2. C	3. C	4. C	5. C
6. B	7. C	8. A	9. B	10. B
11. B	12. C	13. C	14. B	15. C
16. C	17. B	18. B	19. C	20. B
21. A	22. A	23. D	24. A	25. A
26. B	27. D	28. D	29. E	30. B

Section A: Figural Reasoning

Q1: C. Notch now on the TOP

Rotating 270 degrees CLOCKWISE is the same as rotating 90 degrees COUNTER-CLOCKWISE. The notch started on the RIGHT (3 o'clock position). After 270 degrees CW: right goes to top. Option A shows 90 degrees CW (notch at bottom); Option B shows 180 degrees (notch at left); Option D is unchanged; Option E is a diagonal misread.

Q2: C. 24 cubes

Each face of the $4 \times 4 \times 4$ cube has a 4×4 grid of unit faces. The unit cubes with EXACTLY ONE face painted sit in the 2×2 interior of each face (not on the edges or corners). So $2 \times 2 = 4$ such cubes per face, times 6 faces = 24 . Option A (8) is the corner count (3 faces painted). Option B (16) confuses 4×4 with 4×2 .

Q3: C. A large black pentagon

Rows change shape: triangle, square, pentagon. Columns change size+fill together: small white, medium gray, large black. Row 3, column 3 = large black pentagon. Option A keeps the wrong column values. D changes the shape. E mixes size and fill.

Q4: C. 12 cubes

For a $3 \times 3 \times 3$ cube, unit cubes with exactly 2 faces painted sit on the EDGES but not at corners. A cube has 12 edges; each edge has one middle cube (the ones between the two corner cubes on that edge). So 12 edge-middle cubes with 2 faces painted. Corners (8) have 3 faces painted; face centers (6) have 1 face painted; one interior cube (1) has 0 faces painted. Total check: $8 + 12 + 6 + 1 = 27$. Correct.

Q5: C. 486

The rule is geometric: $\text{next} = 3 \times \text{current}$. Check: $2 \times 3 = 6$, $6 \times 3 = 18$, $18 \times 3 = 54$, $54 \times 3 = 162$, $162 \times 3 = 486$. Option A (216) is $162 + 54$ (arithmetic instead of geometric). Option E (648) is 162×4 .

Q6: B. Circle

A plane parallel to the cone's circular base produces a smaller CIRCLE. The cone is a cone, and every horizontal slice is a circular cross-section (similar to the base but smaller). Option C (ellipse) would require the slice to be at an angle. Option A (triangle) would come from slicing vertically through the apex. Option D (parabola) would come from an angled slice parallel to one slant side.

Q7: C. 2 holes, mirrored across the fold line

When paper is folded in half (2 layers) and a hole is punched, the hole passes through both layers. When unfolded, the two holes are symmetric across the fold line. A hole in the upper-left of the folded

piece produces a matching hole in the upper-right of the unfolded sheet. Option A ignores the layering. Option D assumes 4 folds. Option E mis-positions the holes.

Q8: A. Both squares shaded, overlap rectangle empty

"Union minus intersection" is the symmetric difference: the region inside either square but NOT inside both. Visually, this is both squares with the overlap rectangle left empty, producing two L-shaped shaded regions. Option B shows only the intersection. C shows the full union. D shows only one square. Same set-theory concept as Test 05 Q8 but applied to squares instead of circles.

Q9: B. 7.5 degrees

At 3:15, the minute hand points directly at 3 (90 degrees from 12). The hour hand is NOT at 3 exactly; it has moved 1/4 of the way from 3 to 4 (because 15 minutes have elapsed). Each hour mark is 30 degrees, so the hour hand has advanced $30 \times (15/60) = 7.5$ degrees past the 3 position. Angle between: $97.5 - 90 = 7.5$ degrees. Option A (0) is the common wrong answer for students who think both hands are at 3 exactly.

Q10: B. Rectangle (non-square)

A non-square rectangle has 180 degrees rotational symmetry (half-turn looks identical) but NOT 90 degrees (quarter-turn makes width and height swap, which visibly changes the shape). Option A (square) has BOTH 90 and 180 rotational symmetry. Option C (equilateral triangle) has 120 degrees symmetry, not 180. Option D (L-shape) has no rotational symmetry. Option E (circle) has every angle of rotational symmetry.

Section B: Observation with Interference

Q11: B. Vote Forward 2028

The blue backdrop banner had "VOTE FORWARD 2028" in white, with "Senator J. Alvarez" in smaller gold text below. Option C ("Forward Together") is a nearby-sounding slogan; D drops the "Vote Forward" element.

Q12: C. 11:08

The digital clock on the upper-left of the backdrop read 11:08 in red LED.

Q13: C. 3 photographers

Three press photographers (with cameras at chest level) stood in the front row of the crowd. They were spaced at roughly 100, 160, and 220 pixels from the left edge.

Q14: B. VOTE J.A. and FORWARD

Two supporter signs on the right edge of the crowd held by different supporters. The first read "VOTE J.A." (Alvarez's initials), the second read "FORWARD."

Q15: C. J. ALVAREZ

A small plate attached to the front of the podium read "J. ALVAREZ" in small letters. The backdrop above the podium carried the full slogan "Senator J. Alvarez" in gold text; the podium plate uses the shortened form. Option B ("Senator Alvarez") mis-remembers the backdrop and substitutes it for the podium text. Option D ("Vote Forward") conflates the banner slogan with the podium plate.

Q16: C. Washington Monument

The dark blue sign at the base of the obelisk read "WASHINGTON MONUMENT" in gold lettering. The monument's shape (tall tapered obelisk) is also a visual confirmation.

Q17: B. VISITOR ENTRANCE

A green sign in the upper-left corner of Scene 16 read "VISITOR ENTRANCE" in white letters. This is the standard signage at National Mall and monument perimeters. Option C ("EMERGENCY EXIT") is a red-backed sign from Scene 14 (the hospital) in a prior test, making it an interference trap if you have been drilling the whole series.

Q18: B. White shirt with a black vest and gold badge

This is the signature USSS Uniformed Division uniform appearance: white shirt, black tactical vest, gold badge on the vest, and a duty belt. This is DIFFERENT from the plainclothes dark suit worn by

Secret Service Special Agents (seen in prior tests at the hotel and motorcade scenes).

Q19: C. A mounted officer on horseback

On the far right of Scene 16, a mounted USSS officer sat on a brown horse. USSS Uniformed Division operates a Horse Mounted Unit on the White House and monument grounds. Option A (bicycle patrol) and D (golf-cart) are plausible UD support units but were not shown in this scene.

Q20: B. Scene 16, blue

Scene 16 showed a blue reflecting pool in the foreground (the classic Washington Monument grounds feature). Scene 15 was set on a rally stage with no water. Option C is a trap for candidates who remember "water" generically without binding it to the correct scene.

Section C: Logic-Based Reasoning

Q21: A. Officer Park holds a valid TS/SCI clearance

A NECESSARY condition means: if X happens (Park works in SOC), then Y must be true (Park has TS/SCI). By modus ponens on the necessary-condition statement: Park works in SOC, so Park must hold TS/SCI. Option B reverses the direction (making TS/SCI sufficient, not just necessary). Option D confuses necessary with sufficient. This is the classic "only if" direction check.

Q22: A. NORTH: T. EAST: P, R, S. SOUTH: Q

Check all three constraints against option A:

(1) P and Q at different posts: P at EAST, Q at SOUTH. OK.

(2) R and S at same post: both at EAST. OK.

(3) T alone at NORTH: T is the only officer at NORTH. OK.

Option B puts P at NORTH with T, violating constraint 3. Option C puts P and Q both at SOUTH, violating constraint 1. Option D puts P at NORTH instead of T, violating constraint 3. Option E puts P

and Q both at EAST, violating constraint 1.

Q23: D. 24 assignments

Four distinct officers to four distinct posts, one each, is the number of permutations of 4 items: $4! = 4 \times 3 \times 2 \times 1 = 24$. Option A (4) counts officers only. Option C (16) is 4^2 , for 2 posts. Option E (256) is 4^4 , the number if each officer could independently pick any of 4 posts with repeats.

Q24: A. Exactly one of Owens or Park is lying

Owens says "Park is lying." Park says "Owens is lying." This forms a contradiction pair: O and P have OPPOSITE truth values. If O is truthful, then P lies, and P's statement "O lies" is false, which is consistent. If O lies, then P doesn't lie, and P's statement "O lies" is true, also consistent. Either way, EXACTLY ONE of O, P is lying. Nelson and Quinn form a mutual-affirmation pair, so they have the SAME truth value (either both truthful or both lying). Option C (both N and Q truthful) is only sometimes true. Option B is impossible because N and Q cannot have different truth values. Option D and E are impossible because at least one must be lying (the O/P contradiction forces at least one lie).

Q25: A. Officer Lane is certified for carry duty

This is classic modus ponens: "If P then Q. P is true. Therefore Q." Lane completed the course (P), so Lane is certified (Q). Option B contradicts the conclusion. Option D confuses "if then" with "only if." Option E is too cautious; modus ponens gives a definite answer.

Q26: B. The response team did NOT acknowledge

Rule: $(\text{sensor fires AND ack}) \rightarrow \text{log}$. Given NOT log and sensor fired. Modus tollens on the conjunction: $\text{NOT log} \rightarrow \text{NOT (sensor AND ack)} = \text{NOT sensor OR NOT ack}$. Given sensor fired (so NOT sensor is false), only NOT ack can be true. Therefore the response team did NOT acknowledge. Option A contradicts the derivation. Option C contradicts a given fact.

Q27: D. 96 inquiries

Expected value: $12 \text{ inquiries/hour} \times 8 \text{ hours} = 96 \text{ inquiries}$. This is the simplest form of expected frequency: multiply the rate by the duration. Option B (48) uses a 4-hour shift. Option E (120) uses 10 hours.

Q28: D. It cannot be determined

Premises: $K9 \text{ handlers are all UD}$ (subset). $\text{Some UD serve on CS team}$. We know K9 handlers are a subset of UD, and SOME UD are on CS, but we do not know whether any K9 handlers fall within that "some." The K9 handlers might all be non-CS, or some might be CS; the premises do not constrain this. Option B ("some K9 are CS") is a common quantifier trap (the "undistributed middle"). Options A and C over-state the premises.

Q29: E. Saturday

Day 1 is Monday. The day of the week follows a 7-day cycle. Day 20 = Day 1 + 19 days. $19 \bmod 7 = 5$. Monday + 5 = Saturday. The 4-on-2-off schedule is a distraction; the question asks only for the day of the week, which depends on the 7-day cycle, not the duty cycle.

Q30: B. S1 = T, S2 = T, S3 = F, S4 = T

Verify:

S1 = T claims $S2 = T$. S2 is T. Consistent.

S2 = T claims $S1 = T$. S1 is T. Consistent.

S3 = F claims "exactly two of four are true." Actual count: S1, S2, S4 are true (3 true). The claim "exactly 2 true" is FALSE (actual is 3), so S3 being F matches.

S4 = T claims S1 is true. S1 is T. Consistent.

All four statements align under (T, T, F, T). Option A (T,T,T,F) fails: $S3=T$ claims "exactly 2 true" but actual count = 3 (S1, S2, S3), so statement false, $S3=T$ contradicts. Option C (F,F,F,T) fails: $S4=T$ claims S1 is true, but $S1=F$, contradiction. Option D (F,F,T,F) fails: $S3=T$ claims "exactly 2 true" but only S3 itself is true (count = 1), so statement false, $S3=T$ contradicts.

Scoring: Test 08. 27+: exam-ready. **23-26:** solid pass expected. **18-22:** borderline, drill two more weeks. **Below 18:** rebuild from Test 01.

What Test 08 adds over Tests 05-07: painted-cube counting (Q2, Q4) is a UDEE-specific spatial skill; cross-sections (Q6), clock-angle geometry (Q9), and rotational symmetry (Q10) are figural patterns

new to this series. Section C introduces necessary-vs-sufficient reasoning (Q21), permutation counting (Q23), quantifier syllogisms (Q28), and calendar arithmetic (Q29) for the first time. Q30 is the first 4-statement self-referential puzzle in the series; unique consistent assignment is (T, T, F, T).